A Proved Transcript Grammar for the BB(5) Champion

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November 2026

Abstract

The Marxen–Buntrock machine is the 5-state Busy Beaver winner: it halts after 47,176,870 steps. Its high-level behaviour has been known since Buro (1990) and Michel (1993), who showed that it follows a Collatz-like map. We study the same machine through its *transcript*, the sequence of state–symbol pairs it visits, written as decimal digits. We give a closed-form grammar for the entire transcript: it is `0246` followed by 15 layers, each built from a repeated word $(159)^M$, an opening gate (`02` or `1`), a ladder of rungs $4\,7^{n+2}\,6\,0\,3^{n+3}\,2$, a final rung $4\,7^{n+2}\,6\,0\,3^{n+4}\,2$ and the closing word `46`; the last layer ends in `158`. We prove in Coq, with no axioms, that the transcript of the machine from the blank tape is exactly this grammar, and that our layer map is the Buro–Michel map. The step count and the number of ones on the final tape follow from the proved configurations by short exact calculations.

1 Introduction

The Busy Beaver value $S(5) = 47{,}176{,}870$ was determined in 2024 by the bbchallenge collaboration, with a proof checked in Coq [1, 2]. The value is set by the machine found by Marxen and Buntrock in 1989 [3]:

`1RB1LC_1RC1RB_1RD0LE_1LA1LD_1RZ0LA`.

Its long run is well understood at a high level. Buro [4] and, independently, Michel [5] showed that it follows a Collatz-like rule. In Michel's notation, let $g(x)$ be the configuration in state A next to a block of x ones on an otherwise blank tape. Then

$$g(3k) \to g(5k+6), \qquad g(3k+1) \to g(5k+9), \qquad g(3k+2) \to \text{halt}, \tag{1}$$

and from $g(0)$ the orbit $0, 6, 16, 34, \ldots, 12284$ halts after 15 applications [7, 8].

The bbchallenge paper calls the list of state–symbol pairs visited from the blank tape the *transcript* of a machine, and uses back-to-back repeats in the transcript to decide machines that loop forever [1, Section 4.3.1]. The champion halts, so its transcript never repeats with a fixed period. But it does repeat with *growing* counts, and these counts follow (1).

Contributions. This paper describes the champion's transcript exactly.

1. A closed-form grammar for the whole transcript of 47,176,870 symbols (Theorem 1). Each layer is a short word with explicit exponents.

2. A Coq proof, with no axioms, that the transcript of the machine from the blank tape is exactly this grammar (Section 7). As far as we know, earlier machine-checked proofs establish the champion's halting by running it, not by its structure [1, 2].

3. A proof in Coq that our layer map is the Buro–Michel map (1) under the change of variable $a = x + 2$ (Proposition 6).

4. Exact formulas for the steps spent in each layer, which add up to 47,176,870, and a short account of why the final tape holds 4,098 ones (Section 6).

The Collatz-like rule itself is not new. What is new is the transcript-level description and its formal proof.

2 Preliminaries

The machine. The transition table is

	A	B	C	D	E
read 0	1RB	1RC	1RD	1LA	1RZ
read 1	1LC	1RB	0LE	1LD	0LA

The machine starts in state A on a blank tape. We write the halting transition as E0 $\to$ 1RZ; bbchallenge leaves E0 undefined [1]. Both conventions give 47,176,870 steps, and in ours the last step read E0.

Configurations. We write $0^\infty\, u\, [\mathrm{s}\, x]\, v\, 0^\infty$ for the tape $u\, x\, v$ surrounded by blanks, with the head in state s on the cell holding x. Powers denote repetition, e.g. 1^a or $(001)^M$.

Transcripts as digits. The champion has exactly ten state–symbol pairs, so we write each one as a decimal digit:

$$d(s, x) = 2 \cdot \mathrm{idx}(s) + x, \qquad \mathrm{idx(A)} = 0, \ldots, \mathrm{idx(E)} = 4.$$

So A0 = 0, A1 = 1, B0 = 2, ..., E1 = 9. The *transcript* T is the string of digits $d(s_i, x_i)$ for the steps $i = 0, 1, \ldots, 47{,}176{,}869$. This is the bbchallenge transcript [1] written in digits; the digits only make the patterns easier to read.

The first 100 digits. The transcript of the champion begins as follows. Spaces and line breaks are added only to show where the words of the next section start; the digits themselves run on without breaks.

```
0246
159  02  47777603333332  46
159159159  02  4777760333332
47777777776033333333332
4777777777777776033333333333 ...
```

The first line takes the blank tape to the first layer. The second line is the whole first layer: one 159, the opening 02, one rung and the close 46. From the third line on, 159 repeats three times, and each rung is longer than the one before: the runs of 7s grow $4, 9, 14$ and the runs of 3s grow $5, 10, \ldots$

3 The transcript grammar

3.1 Words

We use the following words over the digits 0–9.

Name	Word	Meaning
Ω	159	one oscillation A1, C1, E1
$R(n)$	$4\,7^{n+2}\,6\,0\,3^{n+3}\,2$	internal rung
$F(n)$	$4\,7^{n+2}\,6\,0\,3^{n+4}\,2$	final rung
G_1	02	Gate 1
G_2	1	Gate 2
E	46	close of a layer
H	158	halt

In the notation of the earlier versions, a rung $4\,7^{A}\,6\,0\,3^{B}\,2$ has $B = A + 1$ when internal and $B = A + 2$ when final. We write $K = n + 2$ for the length A of the final rung.

3.2 Layers

A layer starts at the configuration

$$S(a) \;=\; 0^{\infty}\,1^{a}\,[\mathrm{A}\,1]\,1\,0^{\infty},$$

with $a \geq 1$ ones to the left of the head. Write $a + 1 = 3M + r$ with $0 \leq r \leq 2$. The layer word $\mathcal{W}(a)$ and the next start $N(a)$ are:

r	$\mathcal{W}(a)$	$N(a)$
0	$\Omega^{M}\,G_1\,R(2)\,R(7)\cdots R(5M-8)\,F(5M-3)\,E$	$5M + 3$
1	$\Omega^{M}\,G_2\,R(0)\,R(5)\cdots R(5M-5)\,F(5M)\,E$	$5M + 6$
2	$\Omega^{M}\,H$	halt

In words: the layer repeats Ω exactly M times; then the remainder r picks the gate; then the rungs grow by 5 each time; a final rung closes the ladder; and `46` leads into the next layer. A Gate 1 layer has M rungs and a Gate 2 layer has $M + 1$ rungs, counting the final one. The final rung has $K = 5M - 1$ (Gate 1) or $K = 5M + 2$ (Gate 2), and in both cases $N(a) = K + 4$.

Theorem 1 (Transcript grammar)**.** *Let $a_0 = 2$ and $a_{i+1} = N(a_i)$. Then $a_{14} + 1 \equiv 2 \pmod 3$, and the transcript of the champion from the blank tape is exactly*

$$T \;=\; 0246\,\mathcal{W}(a_0)\,\mathcal{W}(a_1)\cdots\mathcal{W}(a_{14}).$$

Proof. Proved in Coq as `bb5_decimal_trace`; see Section 7. The prefix `0246` takes the blank tape to $S(2)$. Lemmas 2–5 below give each layer. □

Table 1 lists the 15 layers. The first two layers read

$$\mathcal{W}(2) = 159\;\;02\;\;4\,7^{4}\,6\,0\,3^{6}\,2\;\;46,$$
$$\mathcal{W}(8) = (159)^{3}\;\;02\;\;4\,7^{4}\,6\,0\,3^{5}\,2\;\;4\,7^{9}\,6\,0\,3^{10}\,2\;\;4\,7^{14}\,6\,0\,3^{16}\,2\;\;46.$$

The prefix `0246` is itself G_1 followed by E: it is a layer with no oscillations and no rungs.

Table 1: The 15 layers of the transcript. a: ones left of the head at the layer start; $x = a - 2$: Michel's value; M: number of Ω; K: length A of the final rung; steps: length of $\mathcal{W}(a)$. The four steps of `0246` come first; the total is 47,176,870.

Layer	a	x	M	Word	Rungs	K	Steps
1	2	0	1	Gate 1	1	4	21
2	8	6	3	Gate 1	3	14	83
3	18	16	6	Gate 2	7	32	295
4	36	34	12	Gate 2	13	62	937
5	66	64	22	Gate 2	23	112	2,807
6	116	114	39	Gate 1	39	194	8,039
7	198	196	66	Gate 2	67	332	22,915
8	336	334	112	Gate 2	113	562	64,637
9	566	564	189	Gate 1	189	944	180,689
10	948	946	316	Gate 2	317	1582	504,665
11	1586	1584	529	Gate 1	529	2644	1,405,029
12	2648	2646	883	Gate 1	883	4414	3,908,163
13	4418	4416	1473	Gate 1	1473	7364	10,864,853
14	7368	7366	2456	Gate 2	2457	12282	30,201,445
15	12286	12284	4095	Halt	–	–	12,288

4 Tape lemmas

Each word of the grammar comes from a simple change on the tape. Each lemma below is proved in Coq for all values of its parameters; the Coq names are given in brackets. The intermediate configurations are close to those used in the analyses of Buro and Michel [4, 5]; we state them because the grammar is read directly from them.

Lemma 2 (Ω block; `omega_block_tr`)**.** *For $M \geq 1$ and any symbol y,*

$$0^\infty\, u\, y\, 1^{3M-1}\, [\mathrm{A}\, 1]\, v\, 0^\infty \xrightarrow{(159)^M} 0^\infty\, u\, [\mathrm{A}\, y]\, (001)^M\, v\, 0^\infty.$$

Each Ω moves the head three cells to the left and turns 111 into 001 (A1 writes 1, C1 and E1 write 0). So M oscillations remove $2M$ ones.

Lemma 3 (Rung; `rung_tr`)**.** *For $n \geq 0$,*

$$0^\infty\, 1^n\, [\mathrm{C}\, 0]\, 1\, 0\, 0\, w \xrightarrow{R(n)} 0^\infty\, 1^{n+5}\, [\mathrm{C}\, 0]\, w,$$

in $2n + 9$ steps, where the head ends on the last 0 of 100.

The rung runs C0, then D sweeps left over $n + 2$ ones (the 7s), D0 and A0 add two ones at the left end (the 6 0), and B sweeps right over $n + 3$ ones (the 3s) before B0 (the 2). One unit 100 is filled, and two cells are added on the left: the block grows by 5. Repeating the lemma gives a ladder $R(n)\, R(n + 5) \cdots$ (`ladder_tr`).

Lemma 4 (Final rung; `terminal_rung_tr`)**.** *For $n \geq 0$,*

$$0^\infty\, 1^n\, [\mathrm{C}\, 0]\, 1\, 1\, 0^\infty \xrightarrow{F(n)} 0^\infty\, 1^{n+6}\, [\mathrm{C}\, 0]\, 0^\infty,$$

in $2n + 10$ steps.

The extra 3 in the B sweep is the trailing pair 11 on the right, which is the leftover of the start configuration $S(a)$.

Lemma 5 (Gates, halt and close)**.** *After the* Ω *block of* $S(a)$*, with* $a+1 = 3M + r$*:*

- $r = 0$*:* $0^\infty\,[\mathrm{A}\,0]\,(001)^M\,1\,0^\infty \xrightarrow{02} 0^\infty\,1\,1\,[\mathrm{C}\,0]\,(100)^{M-1}\,1\,1\,0^\infty$*;*
- $r = 1$*:* $0^\infty\,[\mathrm{A}\,1]\,(001)^M\,1\,0^\infty \xrightarrow{1} 0^\infty\,[\mathrm{C}\,0]\,(100)^M\,1\,1\,0^\infty$*;*
- $r = 2$*:* $0^\infty\,1\,[\mathrm{A}\,1]\,(001)^M\,1\,0^\infty \xrightarrow{158}$ *halt.*

After the final rung, $0^\infty\,1^m\,[\mathrm{C}\,0]\,0^\infty \xrightarrow{46} S(m)$*.*

In Coq these are `layer_gate1_tr`, `layer_gate2_tr`, `layer_halt_tr` and `bridge_46`. The gate is decided by what is left when the Ω block reaches the left end of the block of ones: nothing ($r = 0$), one cell ($r = 1$) or two cells ($r = 2$). With two cells left, E reads a 0 and the machine halts.

Combining the lemmas: in a Gate 1 layer the ladder starts at $n = 2$ and has $M - 1$ internal rungs, so the final rung has $n = 5M - 3$ and the next start is $a' = n + 6 = 5M + 3$. In a Gate 2 layer the ladder starts at $n = 0$ and has M internal rungs, so $n = 5M$ and $a' = 5M + 6$. This is the table for $N(a)$.

5 Relation to the Buro–Michel map

Proposition 6 (`layer_map_is_michel`)**.** *For every* $x \geq 0$*, the layer map* N *at* $a = x + 2$ *is the map* (1) *shifted by* 2*: if* $x = 3k$ *then* $N(x{+}2) = (5k{+}6){+}2$*; if* $x = 3k{+}1$ *then* $N(x{+}2) = (5k{+}9){+}2$*; if* $x = 3k + 2$ *the layer halts.*

So the layer starts a_i are Michel's orbit x_i plus 2 (Table 1). Since $N(a) = K + 4$, Michel's next value is $x' = K + 2$. Two consequences, which restate the known rule in the terms of the grammar:

Corollary 7 (`gate_law`, `recurrence_8_1`)**.** *Let* K *be the final-rung length of a layer. The next layer has*

$$M' = \left\lfloor \tfrac{K+2}{3} \right\rfloor + 1,$$

and it is a Gate 1 layer if $(K + 2) \bmod 3 = 0$*, a Gate 2 layer if* $(K + 2) \bmod 3 = 1$*, and the halting layer if* $(K + 2) \bmod 3 = 2$*.*

For the champion, the remainder 2 first appears after layer 14, where $K + 2 = 12284$.

6 Steps and ones

Steps. The step counts in Lemmas 2–5 give the length of each layer word:

$$|\mathcal{W}(a)| = \begin{cases} 5M^2 + 11M + 5 & \text{Gate 1,} \\ 5M^2 + 17M + 13 & \text{Gate 2,} \\ 3M + 3 & \text{halt.} \end{cases}$$

With the four steps of `0246`, the 15 layers add up to exactly 47,176,870 steps (Table 1). The formulas are sums of the proved step counts; the total is checked by exact integer arithmetic. It is not yet stated inside Coq, because Coq's unary natural numbers make a number of this size impractical; a version with binary numbers is future work. The total agrees with the stage counts of Michel [7], which use different stage boundaries.

Ones. The start $S(a)$ holds $a+2$ ones. Writing $a+1 = 3M + r$ and $g = r+1$, this is $3M+g$, and after the Ω block there are $M+g$ ones (Lemma 2). In the halting layer, $M = 4095$ and $g = 3$, leaving $4095 + 3 = 4098$ ones. The final word 158 does not change this: C1 erases a one and E0 writes one. This gives $\Sigma(5) = 4098$ for the champion.

7 The Coq proof

The development is the file `coq/BB5Omega.v` in the repository (about 840 lines, Coq 8.18). It uses no axioms: `Print Assumptions` on the main theorems reports that they are closed under the global context.

Model. A configuration is a state, the list of cells to the left of the head (nearest first), the cell under the head, and the list of cells to the right. An empty list stands for blanks. `step` applies the transition table, and `trace n c` lists the digits $2 \cdot \mathrm{idx}(s) + x$ of the first n steps from c. The key fact used everywhere is

$$\texttt{trace}\,(p+q)\,c \;=\; \texttt{trace}\,p\,c \;\cdot\; \texttt{trace}\,q\,(\texttt{steps}\,p\,c).$$

Proof structure. Each tape lemma is proved twice: once for the configuration it reaches and once for the digits it writes. Sweeps and blocks of any length are handled by induction. The layer lemmas are put together into `layer_correct_tr`, which covers every $a \geq 1$. A short recursion then runs the layer map from $a = 2$; Coq computes that it halts after 15 layers (`fifteen_layers`) and the sequence of M values (`M_sequence`).

Main statement. Here `grammar 20 2` is the concatenation of the layer words starting from $a = 2$ (20 is an upper bound on the number of layers):

```
Theorem bb5_decimal_trace :
  exists n, state (steps n (A, [], false, [])) = Z
        /\ trace n (A, [], false, []) = [0; 2; 4; 6] ++ grammar 20 2.
```

As an extra check, we computed the first five layers of `grammar` inside Coq (4,147 digits) and compared them with an independent simulation; they agree. A separate Python script in the repository, `verify_gate_law.py`, runs all 47,176,870 steps and checks the layer data of Table 1 from the transcript.

What is not in Coq. The total step count and the final count of ones are derived from the proved lemmas by arithmetic, as in Section 6, but are not stated as Coq theorems.

8 A smaller case: BB(4)

The same lens applied to the 4-state winner `1RB1LB_1LA0LC_1RZ1LD_1RD0RA` (107 steps, 13 ones; digits 0–7) gives

$$0212\ (03567)^3\ 0212\,1\,3567\ (03567)^4\ 0212\,1\,357\,1212\ (03567)^4\ 0212\,1\,357\,12\ (03567)^2\ 0212\,1\,34.$$

There is a repeated word (03567), a gate-like opening (0212) and a halt by a change of one digit at the end of the repeated word ($35 \to 34$, as $159 \to 158$ for BB(5)). But the counts $3, 4, 4, 2$ do not grow, and there is no layer map. This is only an observation on one small run; we do not claim a general pattern.

9 Discussion

Growing repeats. The bbchallenge decider for loops looks for a transcript block that repeats back-to-back with a fixed length [1, Algorithm 1]. The champion's transcript is made of blocks that repeat a growing number of times, and the counts follow a Collatz-like map. A natural question is whether such grammars can be found automatically: detect growing repeats in a transcript, fit a layer map from a few layers, and then iterate the map instead of the machine. For BB(5) the first layers can mislead: $M = 1, 3, 6, 12, 22$ looks like doubling before the ratio settles near 5/3. So several layers, and every remainder class, must be seen before a map is trusted. Even a correct map does not settle halting in general: for machines such as Antihydra [1] the question becomes a hard Collatz-like problem.

Formal proofs of champions. The Coq proof of $S(5)$ needs only that the champion halts within 47,176,870 steps, which it checks by running it [1]. A structural proof, as here, says why. Proofs of this kind may be useful for larger champions, whose runs are far too long to execute.

Future work. (1) State the step count and the count of ones in Coq, using binary numbers. (2) Try transcript grammars on other Collatz-like machines, including BB(6) champions. (3) Extract layer maps from transcripts automatically.

10 Conclusion

The transcript of the BB(5) champion has an exact grammar: `0246` followed by 15 layers of the form $(159)^M$, a gate, a ladder of rungs that grow by five, a final rung and `46`, with a last layer ending in `158`. The grammar is the Buro–Michel map seen at the level of single steps. We proved in Coq, without axioms, that the machine's transcript is exactly this grammar.

Acknowledgements

Parts of the analysis and of the Coq development were carried out with the help of an AI assistant (Claude, Anthropic); the author checked all results, and the Coq proofs are checked by the Coq kernel.

References

[1] The bbchallenge Collaboration: J. Blanchard, D. Briggs, K. Deka, N. Fenner, Y. Forster, G. Georgiev (Skelet), M. L. House, R. Hunter, Iijil, M. Kądziołka, P. Kropitz, S. Ligocki, mxdys, M. Naściszewski, savask, T. Stérin, C. Xu, J. Yuen, T. Zimmermann. Determination of the fifth Busy Beaver value. arXiv:2509.12337, 2025.

[2] mxdys. Coq-BB5: a Coq proof that $S(5) = 47{,}176{,}870$. `https://github.com/ccz181078/Coq-BB5`, 2024.

[3] H. Marxen and J. Buntrock. Attacking the Busy Beaver 5. *Bulletin of the EATCS*, 40:247–251, 1990.

[4] M. Buro. Ein Beitrag zur Bestimmung von Rados $\Sigma(5)$ oder Wie fängt man fleissige Biber? Schriften zur Informatik und angewandten Mathematik, Report No. 146, RWTH Aachen, 1990.

[5] P. Michel. Busy beaver competition and Collatz-like problems. *Archive for Mathematical Logic*, 32:351–367, 1993.

[6] P. Michel. The Busy Beaver competition: a historical survey. arXiv:0906.3749, 2022.

[7] P. Michel. Behavior of busy beavers. `https://bbchallenge.org/~pascal.michel/beh`.

[8] BusyBeaverWiki. BB(5) champion. `https://wiki.bbchallenge.org/wiki/BB(5)_champion`.

[9] S. Aaronson. The Busy Beaver frontier. *SIGACT News*, 51(3):32–54, 2020.

[10] T. Radó. On non-computable functions. *Bell System Technical Journal*, 41(3):877–884, 1962.